

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0293

Covanta Wallingford Energy 100 kV
February 19, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0293. The Interconnection Customer proposes a 252.1 MW wind facility interconnecting at the Covanta Wallingford Energy 100 kV substation in the Wisconsin area. The request was received on November 22, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 199 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,788,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Covanta Wallingford Energy to Zion Energyctr 100 KV
Contingency: Wommack to Santa Teresa 100 KV
Loading: 189.7% (Rating: 192 MVA)
DFAX: 71.2%
Pre-project loading: 96.4%
2. Facility: Zion Energy Center to Army Depot Substation 100 KV Line
Contingency: Santa Teresa to Wommack 100 KV
Loading: 174.4% (Rating: 227 MVA)
DFAX: 71.2%
Pre-project loading: 95.5%
3. Facility: Medco to Covanta Wallingford Energy 100 KV
Contingency: Zion Energy Center to Army Depot 100 KV
Loading: 166.3% (Rating: 152 MVA)
DFAX: 85.1%
Pre-project loading: 25.2%
4. Facility: Prospect Hts to Potter County 100 KV Line
Contingency: Plains to El Grove 100 KV
Loading: 144.7% (Rating: 122 MVA)
DFAX: 16.3%
Pre-project loading: 111.0%
5. Facility: Wommack to Cobined Locks 100 KV
Contingency: Pinhook to Santa Teresa 100 KV
Loading: 137.8% (Rating: 118 MVA)
DFAX: 24.6%
Pre-project loading: 85.4%

6. Facility: Bridgehampton to Quantum Lake Power Lp 100 KV
Contingency: Pinhook to Santa Teresa 100 KV line
Loading: 117.0% (Rating: 133 MVA)
DFAX: 11.7%
Pre-project loading: 94.8%
7. Facility: Crum Canyon to Browns Ferry 100 KV L
Contingency: Santa Teresa to Wommack 100 KV
Loading: 115.4% (Rating: 294 MVA)
DFAX: 28.8%
Pre-project loading: 90.6%
8. Facility: Chesterfield to Army Depot 100 KV
Contingency: Salinas to Army Depot 100 KV
Loading: 111.8% (Rating: 215 MVA)
DFAX: 61.4%
Pre-project loading: 39.9%
9. Facility: E Substation to Potter County 100 KV
Contingency: Dell Substation to Potter County 100 KV
Loading: 110.3% (Rating: 127 MVA)
DFAX: 10.1%
Pre-project loading: 90.1%
10. Facility: Browns Ferry to Pinhook 100 KV
Contingency: Wommack to Santa Teresa 100 KV
Loading: 108.2% (Rating: 358 MVA)
DFAX: 28.8%
Pre-project loading: 87.9%
11. Facility: Wommack to Santa Teresa 100 KV
Contingency: Santa Teresa to Pinhook Substation 100 KV
Loading: 107.5% (Rating: 407 MVA)
DFAX: 28.6%
Pre-project loading: 89.8%
12. Facility: Santa Teresa to Pinhook 100 KV L
Contingency: Santa Teresa to Wommack 100 KV
Loading: 105.8% (Rating: 387 MVA)
DFAX: 28.8%
Pre-project loading: 87.0%
13. Facility: Simpson Stuart Tap to Fiskill Plains 345 KV
Contingency: Medora to Hampton Rd 345 KV
Loading: 102.8% (Rating: 1,284 MVA)
DFAX: 6.4%
Pre-project loading: 101.5%
14. Facility: Salinas to Army Depot 100 KV
Contingency: Chesterfield to Army Depot 100 KV
Loading: 102.4% (Rating: 224 MVA)
DFAX: 50.5%
Pre-project loading: 45.7%
15. Facility: Dell Substation to Potter County 100 KV
Contingency: East to Dell 100 KV
Loading: 101.7% (Rating: 150 MVA)
DFAX: 9.6%
Pre-project loading: 85.6%
16. Facility: E Ridge to Combined Locks 100 KV line
Contingency: Combined Locks Substation to Wommack Substation 100 KV Line

Loading: 101.4% (Rating: 122 MVA)

DFAX: 27.3%

Pre-project loading: 44.8%

17. Facility: Sunol Tap to Rio Bravo Rocklin Substation 100 KV L

Contingency: Texon to E Bismarck 230 KV

Loading: 101.3% (Rating: 174 MVA)

DFAX: 6.5%

Pre-project loading: 91.9%

18. Facility: East to Dell 100 KV

Contingency: Potter County to Dell 100 KV

Loading: 101.2% (Rating: 161 MVA)

DFAX: 10.1%

Pre-project loading: 85.4%

19. Facility: Elk Grove to Plains 100 KV

Contingency: Estil Springs 230/1kV Autotransformer

Loading: 100.7% (Rating: 259 MVA)

DFAX: 12.6%

Pre-project loading: 88.5%

20. Facility: Labelle to Salinas 100 KV

Contingency: Chesterfield to Army Depot 100 KV line

Loading: 100.6% (Rating: 205 MVA)

DFAX: 50.5%

Pre-project loading: 38.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Covanta Wallingford Energy to Zion Energyctr 100 KV	Rebuild line with higher-capacity conductor	\$3,639,000	48% (\$1,735,000)
Army Depot to Zion Energy Ctr 100 KV Line	Rebuild line with higher-capacity conductor	\$4,807,000	48% (\$2,291,000)
Medco to Covanta Wallingford Energy 100 KV	Rebuild line with higher-capacity conductor	\$2,168,000	100% (\$2,168,000)
Prospect Heights to Potter County 100 KV	Rebuild line with higher-capacity conductor	\$1,812,000	38% (\$682,000)
Wommack to Combined Locks 100 KV Line	Rebuild line with higher-capacity conductor	\$1,818,000	30% (\$536,000)
Bridgehampton to Quantum Lake Power Lp 100 KV	Rebuild line with higher-capacity conductor	\$3,423,000	33% (\$1,119,000)
Crum Canyon to Browns Ferry 100 KV L	Rebuild line with higher-capacity conductor	\$3,799,000	19% (\$724,000)
Army Depot to Cesterfield 100 KV Line	Rebuild line with higher-capacity conductor	\$3,348,000	62% (\$2,066,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E to Potter County 100 KV line	Rebuild line with higher-capacity conductor	\$902,000	38% (\$345,000)
Browns Ferry to Pinhook 100 KV	Reconductor line	\$611,000	19% (\$116,000)
Wommack to Santa Teresa 100 KV line	Reconductor line	\$5,396,000	17% (\$930,000)
Santa Teresa to Pinhook 100 KV L	Reconductor line	\$1,729,000	19% (\$329,000)
Simpson Stuart Tap to Fiskill Plains 345 KV	Reconductor line	\$14,527,000	3% (\$406,000)
Salinas to Army Depot 100 KV	Reconductor line	\$705,000	63% (\$442,000)
Dell to Potter County 100 KV line	Reconductor line	\$939,000	38% (\$359,000)
E Ridge to Combined Locks 100 KV	Reconductor line	\$998,000	39% (\$390,000)
Sunol Tap to Rio Bravo Rocklin Substation 100 KV L	Reconductor line	\$1,327,000	11% (\$149,000)
East to Dell 100 KV	Reconductor line	\$492,000	38% (\$188,000)
Plains to Elk Grove 100 KV	Reconductor line	\$1,067,000	36% (\$389,000)
Labelle to Salinas 100 KV	Reconductor line	\$675,000	63% (\$424,000)
	Total allocated to project		\$15,788,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.